

Towards a Crisp Representation of Fuzzy Description Logics under Łukasiewicz Semantics

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- 3 The Fuzzy DL *ALCHOI*
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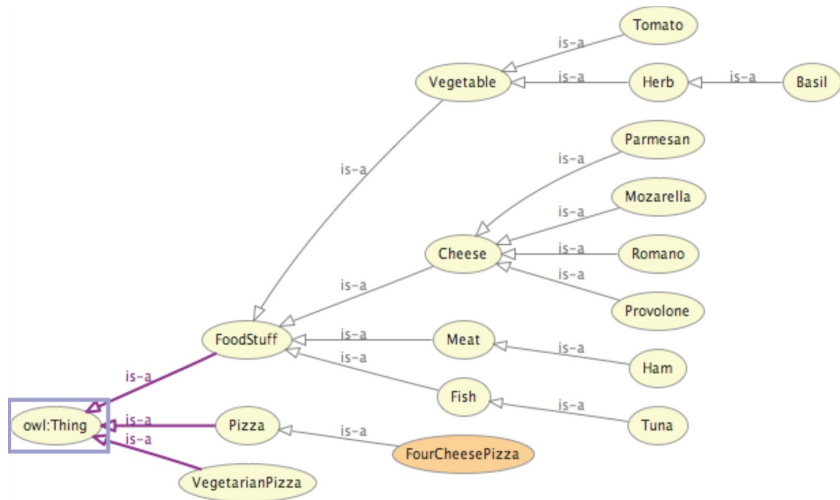


Description Logics

- **Description Logics** (DLs) are a family of logics for representing structured knowledge.
- They describe a domain in terms of concepts (classes), roles (properties, relationships) and individuals.
 - Complex concepts and roles can be built.
- Distinguished by a formal semantics and by providing some inference services: consistency, subsumption . . .
- More popularity due to their application in **Semantic Web**, as the theoretical counterpart of the standard language for ontology representation OWL (Web Ontology Language).
 - OWL DL is the highest level such that reasoning is decidable.
- Each logic is denoted by using a string of capital letters which identify the constructs of the logic and therefore its complexity.
 - For instance, *ALCHOI* is a subset of *SHOIN(D)*, which is the subjacent logic of OWL DL language.



Ontology in a Description-Logic Based Language



- DLs are not appropriate for **fuzzy/vague/imprecise** knowledge for which a clear and precise definition is not possible.
 - An inn is a **cheap** and **small** hotel.
 - Patient001's Serotonin Level is **quite low**.
 - English is **generally** spoken in Canada.
 - I do not like flamenco **very much**.
- **Fuzzy logic** and fuzzy set theory are well known formalisms for representing these types of knowledge.
- **Fuzzy DLs** extend DLs with fuzzy logic.



Crisp Representations for Fuzzy DLs

- Fuzzy DLs require that new languages need to be used, and hence to adapt the available resources.
 - Specially difficult with reasoners: significant gap between the design of a decision procedure and a practical implementation.
- Alternative: To **represent fuzzy DLs using crisp DLs** and to reduce reasoning within fuzzy DLs to reasoning within crisp ones.
- Advantages:
 - Use of standard languages and reuse of available resources.
 - Use of existing crisp reasoners.
 - This will support early adoption of future fuzzy languages.
 - Currently, the only implemented reasoner for fuzzy OWL DL (**DeLorean**) is based on a reduction to a crisp DL.
- An immediate practical application of fuzzy ontologies is feasible, because it relies on existing valid languages and tools.



- Although there has been a relatively significant amount of work in extending DLs with fuzzy logic, the representation of them using crisp DLs has not received such attention.
- Previous work in crisp representation of fuzzy DLs has been restricted to the Zadeh family of fuzzy operators, which has some limitations.
- This work generalizes previous work and provides a crisp **representation of fuzzy *ALCHOT*** under **Łukasiewicz** semantics.

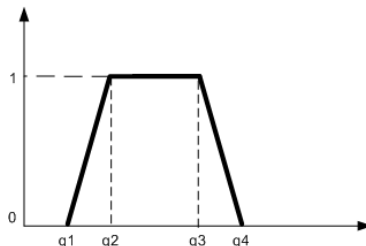


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Fuzzy logic

- Fuzzy set theory and fuzzy logic (Zadeh, 1965) manage imprecise and vague knowledge.
- In classical set theory elements either belong to a set or not.
- In fuzzy set theory elements can belong to **some degree in $[0, 1]$** .
 - 0 means no-membership.
 - 1 means full membership.
 - A value between 0 and 1 represents the extent to which an element can be considered as an element of the fuzzy set.
- **α -cut**: Set of elements that belong to a fuzzy set with degree α .
- Example: **Trapezoidal** membership function of a fuzzy set:



Families of fuzzy logics

- All crisp set **operations** are extended to fuzzy sets.
 - Intersection: t-norm function.
 - Union: t-conorm function.
 - Complement: negation function.
 - Implication: implication function.
- Fuzzy operators are grouped in **families**

Family	t-norm $\alpha \otimes \beta$	t-conorm $\alpha \oplus \beta$	negation $\ominus \alpha$	implication $\alpha \Rightarrow \beta$
Zadeh	$\min\{\alpha, \beta\}$	$\max\{\alpha, \beta\}$	$1 - \alpha$	$\max\{1 - \alpha, \beta\}$
Łukasiewicz	$\max\{\alpha + \beta - 1, 0\}$	$\min\{\alpha + \beta, 1\}$	$1 - \alpha$	$\min\{1 - \alpha + \beta, 1\}$
Gödel	$\min\{\alpha, \beta\}$	$\max\{\alpha, \beta\}$	$\begin{cases} 1, & \alpha = 0 \\ 0, & \alpha > 0 \end{cases}$	$\begin{cases} 1 & \alpha \leq \beta \\ \beta, & \alpha > \beta \end{cases}$
Product	$\alpha \cdot \beta$	$\alpha + \beta - \alpha \cdot \beta$	$\begin{cases} 1, & \alpha = 0 \\ 0, & \alpha > 0 \end{cases}$	$\begin{cases} 1 & \alpha \leq \beta \\ \beta/\alpha, & \alpha > \beta \end{cases}$

- Zadeh family can be represented using Łukasiewicz:
 - negation $\neg$: $\ominus \alpha$
 - t-norm $\wedge$: $\alpha \otimes (\alpha \Rightarrow \beta)$
 - t-conorm $\vee$: $\neg((\neg \alpha) \wedge (\neg \beta))$
 - Implication: $(\neg \alpha) \vee \beta$



Why to use Łukasiewicz logic?

- Łukasiewicz logic is **more general** than Zadeh family.
- In some applications, Zadeh family is not enough. Example:
Fuzzy matchmaking.
 - A concept **Buy** collects all the **buyer's preferences** together in such a way that the higher is the maximal degree of satisfiability, the more the buyer is satisfied, e.g.,

$$\text{Buy} \equiv \text{Sedan} \sqcap (\exists \text{hasPrice.about}30000) \sqcap (\exists \text{hasColor.Black})$$

- A concept **Sell** collects all the **seller's preferences** together in such a way that the higher is the maximal degree of satisfiability, the more the seller is satisfied.
- The **best agreement** between them is determined by the maximal degree of satisfiability of the conjunction $\text{Buy} \sqcap \text{Sell}$.
- Using Łukasiewicz t-norm the solution is **Pareto optimal**.
- But this does not hold for the minimum t-norm (Zadeh family)!!



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- Vocabulary of the language:
 - **Individuals**: fernando.
 - **Concepts**, fuzzy sets of individuals: Tall.
 - **Roles**, fuzzy binary relations over individuals: isFriendOf.
- **Complex definitions** of concepts and roles can be built.
- A **fuzzy interpretation** $\mathcal{I}$ is a pair $(\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$.
 - $\Delta^{\mathcal{I}}$ is a non empty set, the interpretation domain
 - $\cdot^{\mathcal{I}}$ is an interpretation function mapping:
 - Each concept onto a function $C^{\mathcal{I}} : \Delta^{\mathcal{I}} \rightarrow [0, 1]$
 - Each role onto a function $R^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \rightarrow [0, 1]$
- A **fuzzy knowledge base** consists of fuzzy axioms organized in:
 - A fuzzy **ABox** $\mathcal{A}$: extensional knowledge about individuals.
 - A fuzzy **TBox** $\mathcal{T}$: intensional knowledge about concepts.
 - A fuzzy **RBox** $\mathcal{R}$: intensional knowledge about roles.
- Most reasoning tasks are reducible to **KB consistency**.



Fuzzy concepts and roles

Constructor	Syntax	Semantics
top	$\top$	1
bottom	$\perp$	0
atomic concept	Blond	$A^{\mathcal{I}}(a)$
concept conjunction	Human $\sqcap$ Blond	$C^{\mathcal{I}}(a) \otimes D^{\mathcal{I}}(a)$
concept disjunction	Clever $\sqcup$ Blond	$C^{\mathcal{I}}(a) \oplus D^{\mathcal{I}}(a)$
concept negation	$\neg$ Blond	$\ominus C^{\mathcal{I}}(a)$
universal quantification	$\forall \text{hasChild. Human}$	$\inf_{b \in \Delta^{\mathcal{I}}} \{R^{\mathcal{I}}(a, b) \Rightarrow C^{\mathcal{I}}(b)\}$
existential quantification	$\exists \text{hasChild. Woman}$	$\sup_{b \in \Delta^{\mathcal{I}}} \{R^{\mathcal{I}}(a, b) \otimes C^{\mathcal{I}}(b)\}$
fuzzy nominals	$\{1/\text{germany}, 1/\text{austria}, 0.67/\text{switzerland}\}$	$\sup_{i \mid a \in \{o_i^{\mathcal{I}}\}} \alpha_i$
atomic role	hasChild	$R_A^{\mathcal{I}}(a, b)$
inverse role	hasParent ⁻	$R^{\mathcal{I}}(b, a)$



Axiom	Syntax	Semantics
concept assertion	$\langle \text{fernando} : \text{Human} \geq \alpha \rangle$	$C^{\mathcal{I}}(a^{\mathcal{I}}) \geq \alpha$
concept assertion	$\langle \text{fernando} : \text{Woman} \leq \beta \rangle$	$C^{\mathcal{I}}(a^{\mathcal{I}}) \leq \beta$
role assertion	$\langle (\text{fernando}, \text{umberto}) : \text{isFriendOf} \geq \alpha \rangle$	$R^{\mathcal{I}}(a^{\mathcal{I}}, b^{\mathcal{I}}) \geq \alpha$
general concept inclusion	$\langle \text{Inn} \sqsubseteq \text{Hotel} \geq \alpha \rangle$	$\inf_{a \in \Delta^{\mathcal{I}}} \{C^{\mathcal{I}}(a) \Rightarrow D^{\mathcal{I}}(a)\} \geq \alpha$
role hierarchy	$\langle \text{isComponentOf} \sqsubseteq \text{isPartOf} \geq \alpha \rangle$	$\inf_{a, b \in \Delta^{\mathcal{I}}} \{R^{\mathcal{I}}(a, b) \Rightarrow R'^{\mathcal{I}}(a, b)\} \geq \alpha$

- Negative role assertions, GCIs or RIAs (of the form $\tau \leq \beta$) are not allowed, because they correspond to negated axioms, which are not part of the crisp DL.
- We are only considering inequalities of the form $\geq$ and $\leq$, but extending the language with $>$ and $<$ is not complicated.



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Idea of the reduction

- 1 Compute the **set of degrees of truth** which must be considered:
Degrees γ in the fuzzy KB and their complementaries $1 - \gamma$.
 - $\neg \gamma$ is $1 - \gamma$.
 - $\gamma_1 \otimes \gamma_2$ and $\gamma_1 \oplus \gamma_2$ are γ_1 or γ_2 .
 - $\gamma_1 \Rightarrow \gamma_2$ is $1 - \gamma_1$ or γ_2 .
- 2 For each fuzzy atomic concept and role, add **new crisp elements** (α -cut and strict α -cut).
- 3 Add **new axioms** to preserve their semantics.
 - $Hotel_{\geq 0.25} \sqsubseteq Hotel_{>0}, Hotel_{>0.25} \sqsubseteq Hotel_{\geq 0.25}, \dots$
 - $IsIn_{\geq 0.25} \sqsubseteq IsIn_{>0}, IsIn_{>0.25} \sqsubseteq IsIn_{\geq 0.25}, \dots$
- 4 Reduce **fuzzy axioms** using the new crisp elements.
 - The size of the crisp KB is **quadratic**.
 - The size is linear under a fixed set of degrees.
 - The reduction **preserves reasoning**.
 - Consistency of the fuzzy KB and the crisp KB are equivalent.



Example: Reduction of a concept conjunction

- Let us consider the axiom: $\langle a : C \sqcap D \geq \alpha \rangle$.
- If it is satisfied, there exists an interpretation $\mathcal{I}$ such that

$$\min\{C^{\mathcal{I}}(a), D^{\mathcal{I}}(a)\} \geq \alpha$$

- Hence, we know that $C^{\mathcal{I}}(a) \geq \alpha$ and that $D^{\mathcal{I}}(a) \geq \alpha$.
- This is equivalent to the crisp axiom: $a : \rho(C, \geq \alpha) \sqcap \rho(D, \geq \alpha)$.
- $\rho(C, \geq \alpha)$ is the α -cut of C , i.e. the crisp set of elements that belong to C with a degree greater or equal than α .



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Fixed number of degrees

- The reduction for Zadeh family is based on the fact that minimum, maximum, negation and implication do not introduce new degrees.
- However, this is not true in Łukasiewicz family. Example:
 $0.5 \otimes 0.7 = \max\{0.5 + 0.7 - 1, 0\} = 0.2$.
- Solution: **To restrict the set of degrees of truth** in the KB to $\mathcal{N} = \{0, \frac{1}{q}, \frac{2}{q}, \dots, \frac{(q-1)}{q}, 1\}$.
- Let $\frac{a}{q}, \frac{b}{q} \in \mathcal{N}$. In Łukasiewicz logic, $\ominus \frac{a}{q}, \frac{a}{q} \otimes \frac{b}{q}, \frac{a}{q} \oplus \frac{b}{q}, \frac{a}{q} \Rightarrow \frac{b}{q} \in \mathcal{N}$.
 - $\ominus \frac{a}{q} = 1 - \frac{a}{q} = \frac{q-a}{q} \in \mathcal{N}$.
 - $\frac{a}{q} \otimes \frac{b}{q} = \max\{\frac{a}{q} + \frac{b}{q} - 1, 0\} \in \mathcal{N}$.
 - $\frac{a}{q} \oplus \frac{b}{q} = \min\{\frac{a}{q} + \frac{b}{q}, 1\} \in \mathcal{N}$.
 - $\frac{a}{q} \Rightarrow \frac{b}{q} = \min\{1 - \frac{a}{q} + \frac{b}{q}, 1\} \in \mathcal{N}$.



Example: Reduction of a concept conjunction

- Consider $\langle a : A_1 \sqcap A_2 \geq 0.5 \rangle$ and $\mathcal{N} = \{0, 0.25, 0.5, 0.75, 1\}$.
- Every model satisfies $\max\{A_1^{\mathcal{I}}(a) + A_2^{\mathcal{I}}(a) - 1, 0\} \geq 0.5$.
- Hence, $A_1^{\mathcal{I}}(a) + A_2^{\mathcal{I}}(a) - 1 \geq 0.5 \Leftrightarrow A_1^{\mathcal{I}}(a) + A_2^{\mathcal{I}}(a) \geq 1.5$.
- We do not know exactly the truth degrees of $A_1^{\mathcal{I}}(a)$ and $A_2^{\mathcal{I}}(a)$, but we know that they belong to $\mathcal{N}$, so there are 6 possibilities:

$A_1^{\mathcal{I}}(a)$	$A_2^{\mathcal{I}}(a)$	$(A_1 \sqcap A_2)^{\mathcal{I}}(a)$
0.5	1	0.5
0.75	0.75	0.5
0.75	1	0.75
1	0.5	0.5
1	0.75	0.75
1	1	1

- Hence: $a : \{(A_{1 \geq 0.5} \sqcap A_{2 \geq 1}) \sqcup (A_{1 \geq 0.75} \sqcap A_{2 \geq 0.75}) \sqcup (A_{1 \geq 0.75} \sqcap A_{2 \geq 1}) \sqcup (A_{1 \geq 1} \sqcap A_{2 \geq 0.5}) \sqcup (A_{1 \geq 1} \sqcap A_{2 \geq 0.75}) \sqcup (A_{1 \geq 1} \sqcap A_{2 \geq 1})\} \equiv$
 $a : (A_{1 \geq 0.5} \sqcap A_{2 \geq 1}) \sqcup (A_{1 \geq 0.75} \sqcap A_{2 \geq 0.75}) \sqcup (A_{1 \geq 1} \sqcap A_{2 \geq 0.5})$.



Reduction of the fuzzy axioms

- **Fuzzy axioms** are reduced as follows:

Fuzzy axiom	Reduction
$\sigma(\langle a: C \geq \alpha \rangle)$	$a: \rho(C, \geq \alpha)$
$\sigma(\langle a: C \leq \beta \rangle)$	$a: \rho(C, \leq \beta)$
$\sigma(\langle (a, b): R \geq \alpha \rangle)$	$(a, b): \rho(R, \geq \alpha)$
$\kappa(\langle C \sqsubseteq D \geq \alpha \rangle)$	$\bigcup_{\gamma_1, \gamma_2} \{ \rho(C, \geq \gamma_1) \sqsubseteq \rho(D, \geq \gamma_2) \}$ for every pair $\langle \gamma_1, \gamma_2 \rangle$ such that $\gamma_1, \gamma_2 \in \mathcal{N}^+$ and $\gamma_1 = \gamma_2 + 1 - \alpha$
$\kappa(\langle R \sqsubseteq R' \geq \alpha \rangle)$	$\bigcup_{\gamma_1, \gamma_2} \{ \rho(R, \geq \gamma_1) \sqsubseteq \rho(R', \geq \gamma_2) \}$ for every pair $\langle \gamma_1, \gamma_2 \rangle$ such that $\gamma_1, \gamma_2 \in \mathcal{N}^+$ and $\gamma_1 = \gamma_2 + 1 - \alpha$

- **Fuzzy concept and roles** are mapped using a mapping ρ , which is inductively defined as shown in the next slide.
- For instance, given a fuzzy concept C , $\rho(C, \geq \alpha)$ is the crisp set of elements that belong to C with a degree greater or equal α .



Mapping of concept and role expressions

x	y	$\rho(x, y)$
$\top$	$\geq \alpha$	$\top$
$\top$	$\leq \beta$	$\perp$
$\perp$	$\geq \alpha$	$\perp$
$\perp$	$\leq \beta$	$\top$
A	$\geq \alpha$	$A_{\geq \alpha}$
A	$\leq \beta$	$\neg A_{> \gamma}$
R_A	$\geq \alpha$	$R_{\geq \alpha}$
$\neg C$	$\bowtie \gamma$	$\rho(C, \bowtie 1 - \gamma)$
$C \sqcap D$	$\geq \alpha$	$\sqcup_{\gamma_1, \gamma_2} \rho(C, \geq \gamma_1) \sqcap \rho(D, \geq \gamma_2)$ for every pair $\langle \gamma_1, \gamma_2 \rangle$ such that $\gamma_1, \gamma_2 \in \mathcal{N}^+$, $\gamma_1 + \gamma_2 = 1 + \alpha$
$C \sqcap D$	$\leq \beta$	$\rho(\neg C, \geq 1 - \beta) \sqcup \rho(\neg D, \geq 1 - \beta)$
$C \sqcup D$	$\geq \alpha$	$\rho(C, \geq \alpha) \sqcup \rho(D, \geq \alpha) \sqcup_{\gamma_1, \gamma_2} \rho(C, \geq \gamma_1) \sqcap \rho(D, \geq \gamma_2)$ for every pair $\langle \gamma_1, \gamma_2 \rangle$ such that $\gamma_1, \gamma_2 \in \mathcal{N}^+$, $\gamma_1 + \gamma_2 = \alpha$
$C \sqcup D$	$\leq \beta$	$\rho(\neg C, \geq 1 - \beta) \sqcap \rho(\neg D, \geq 1 - \beta)$
$\exists R.C$	$\geq \alpha$	$\sqcup_{\gamma_1, \gamma_2} \exists \rho(R, \geq \gamma_1). \rho(C, \geq \gamma_2)$ for every pair $\langle \gamma_1, \gamma_2 \rangle$ such that $\gamma_1, \gamma_2 \in \mathcal{N}^+$, $\gamma_1 + \gamma_2 = 1 + \alpha$
$\exists R.C$	$\leq \beta$	$\rho(\forall R. \neg C, \geq 1 - \beta)$
$\forall R.C$	$\geq \alpha$	$\sqcap_{\gamma_1, \gamma_2} \forall \rho(R, \geq \gamma_1). \rho(C, \geq \gamma_2)$ for every pair $\langle \gamma_1, \gamma_2 \rangle$ such that $\gamma_1, \gamma_2 \in \mathcal{N}^+$ and $\gamma_1 = \gamma_2 + 1 - \alpha$
$\forall R.C$	$\leq \beta$	$\rho(\exists R. \neg C, \geq 1 - \beta)$



Example: Reduction of a fuzzy KB

- Let $\mathcal{N} = \{0, 0.25, 0.5, 0.75, 1\}$ and $\mathcal{K}$ contain these axioms:
 - $\langle a : \forall R.(C \sqcap D) \geq 0.75 \rangle$,
 - $\langle (a, b) : R \geq 0.75 \rangle$,
 - $\langle b : \neg C \geq 0.75 \rangle$.
- $\mathcal{K}$ is clearly **unsatisfiable**:
 - From the third assertion it follows that $C^{\mathcal{I}}(b^{\mathcal{I}}) \leq 0.25$.
 - Hence $(C \sqcap D)^{\mathcal{I}}(b^{\mathcal{I}}) = \max\{C^{\mathcal{I}}(b^{\mathcal{I}}) + D^{\mathcal{I}}(b^{\mathcal{I}}) - 1, 0\} \leq 0.25$.
 - But from the two former assertions, it follows that $(C \sqcap D)^{\mathcal{I}}(b^{\mathcal{I}}) \geq 0.75 \otimes 0.75 = 0.5$, which is a contradiction.
- Next, we create some new crisp atomic concepts and roles, as well as some **new axioms** to preserve their semantics:
 - $C_{\geq 1} \sqsubseteq C_{>0.75}, C_{>0.75} \sqsubseteq C_{\geq 0.75}, C_{\geq 0.75} \sqsubseteq C_{>0.5}, C_{>0.5} \sqsubseteq C_{\geq 0.5},$
 $C_{\geq 0.5} \sqsubseteq C_{>0.25}, C_{>0.25} \sqsubseteq C_{\geq 0.25}, C_{\geq 0.25} \sqsubseteq C_{>0}, D_{\geq 1} \sqsubseteq D_{>0.75}, \dots$
 - $R_{\geq 1} \sqsubseteq R_{\geq 0.75}, R_{\geq 0.75} \sqsubseteq R_{\geq 0.5}, R_{\geq 0.5} \sqsubseteq R_{\geq 0.25}.$



Example: Reduction of a fuzzy KB

- Now we compute the **reduction of the fuzzy axioms** in $\mathcal{K}$:
 - $\sigma(\langle (a, b) : R \geq 0.75 \rangle) = (a, b) : \rho(R, \geq 0.75) = (a, b) : R_{\geq 0.75}$.
 - $\sigma(\langle b : \neg C \geq 0.75 \rangle) = b : \rho(\neg C, \geq 0.75) = b : \neg C_{>0.25}$.
 - $\sigma(\langle a : \forall R.(C \sqcap D) \geq 0.75 \rangle) = a : \rho(\forall R.(C \sqcap D), \geq 0.75) =$
 $a : [\forall \rho(R, \geq 0.5). \rho(C \sqcap D, \geq 0.25) \sqcap \forall \rho(R, \geq 0.75). \rho(C \sqcap D, \geq 0.5)]$
 $\sqcap \forall \rho(R, \geq 1). \rho(C \sqcap D, \geq 0.75)]$, where:
 - $\rho(R, \geq 0.5) = R_{\geq 0.5}$,
 - $\rho(R, \geq 0.75) = R_{\geq 0.75}$,
 - $\rho(R, \geq 1) = R_{\geq 1}$,
 - $\rho(C \sqcap D, \geq 0.25) =$
 $(C_{\geq 0.25} \sqcap D_{\geq 1}) \sqcup (C_{\geq 0.5} \sqcap D_{\geq 0.75}) \sqcup (C_{\geq 0.75} \sqcap D_{\geq 0.5}) \sqcup (C_{\geq 1} \sqcap D_{\geq 0.25})$,
 - $\rho(C \sqcap D, \geq 0.5) = (C_{\geq 0.5} \sqcap D_{\geq 1}) \sqcup (C_{\geq 0.75} \sqcap D_{\geq 0.75}) \sqcup (C_{\geq 1} \sqcap D_{\geq 0.5})$,
 - $\rho(C \sqcap D, \geq 0.75) = (C_{\geq 0.75} \sqcap D_{\geq 1}) \sqcup (C_{\geq 1} \sqcap D_{\geq 0.75})$,
- It can be seen that **the resulting KB is unsatisfiable**.



Complexity of the reduction

Theorem

Assuming a truth space $\{0, \frac{1}{q}, \frac{2}{q}, \dots, \frac{(q-1)}{q}, 1\}$, a $\mathcal{L}$ -ALCHOI fuzzy KB $\mathcal{K}$ is satisfiable iff its crisp reduction is satisfiable.

- The size of the resulting KB is $\mathcal{O}(n^k)$, where k is the maximal depth of the concepts appearing in the fuzzy KB.
- The depth of a concept is inductively defined:
 - $depth(A) = depth(\neg A) = depth(\top) = depth(\perp) = 1$,
 - $depth(\forall R.C) = depth(\exists R.C) = 1 + depth(C)$,
 - $depth(C \sqcap D) = depth(C \sqcup D) = 1 + \max\{depth(C), depth(D)\}$.
- Under Zadeh semantics, the size of the resulting KB is quadratic (linear under a fixed set of truth degrees).
- But now, we cannot infer the truth degrees, but we need to build **disjunctions or conjunctions over all possible degrees**.



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● Conclusions:

- We have provided a crisp representation for fuzzy *ALCHOI* under Łukasiewicz semantics.
- We need to assume a fixed set of allowed degrees of truth.
- We generalize previous approaches which consider Zadeh semantics.
- From a practical point of view, the size of the resulting KB is much more complex in this case, so the practical feasibility of this approach has to be empirically verified.
- The idea behind our reduction is modular and could be applied to more expressive DLs.

● Future work:

- Extend the expressiveness of the logic.
 - Support transitive roles and cardinality restrictions, reaching fuzzy *SHOIN* (and hence OWL DL).
- Implementation of the proposed reduction.
- Study possible optimizations in some particular common situations.



Questions?

Thank you very much for your attention

